

Adopting the NIST Cybersecurity Framework

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What is the NIST Cybersecurity Framework?

- ▶ Created for organizations of all sizes and sectors to manage a variety of risk factors
- ▶ Designed to guide organizations to understand and select specific outcomes based on unique needs
- ▶ Components include: CSF Core, CSF Organizational Profiles, and CSF Tiers

Govern

- ▶ Sets baseline for how to achieve all other functions
- ▶ Addresses overall strategy, roles, responsibilities, policy, and oversight in the context of organization



Identify, Protect, Detect



- ▶ **Identify:** Determine organizations assets that need protecting.
- ▶ **Protect:** Provide the proper cybersecurity tools to support identified assets.
- ▶ **Detect:** Establish method for detection of threats in a timely and effective manner.

Respond and Recover

- ▶ Respond: Actions required to contain the detected threat.
- ▶ Recover: Plan for returning to normal functions after threat has been mitigated.



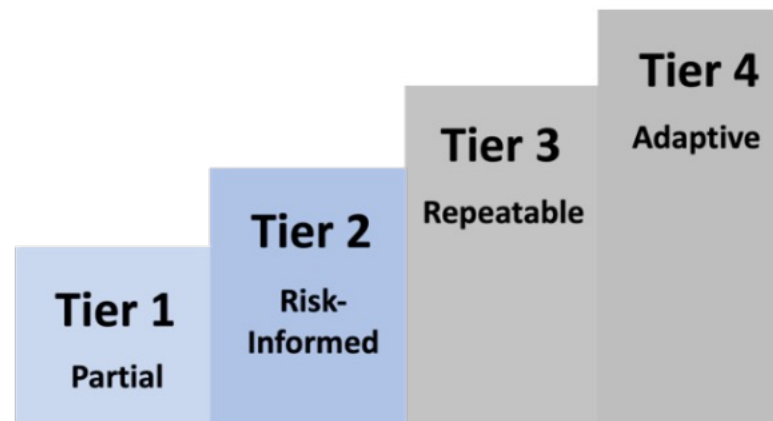
CSF Profiles

- ▶ Current Profile: Core outcomes organization is already achieving
- ▶ Target Profile: Desired outcomes based on organization's cybersecurity risk management objectives



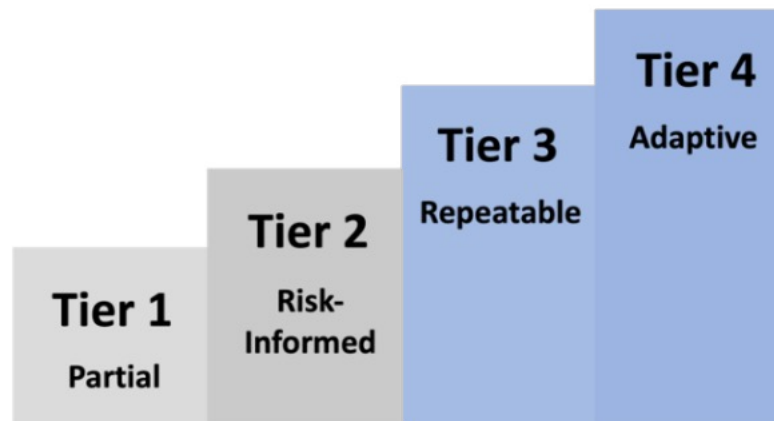
CSF Tiers

- ▶ Tier One: Risk is managed on a case-by-case basis with no formal objectives or knowledge of the threat landscape.
- ▶ Tier Two: Risk management is approved by leadership in the form of understanding risk objectives, threat environments, and business requirements, but not adopted throughout the organization.



CSF Tiers (cont.)

- ▶ Tier Three: Risk management is a formally approved policy and updated with risks communicated throughout the organization.
- ▶ Tier Four: Risk management includes policies, processes, and procedures, is continuously monitored, and adapted based on current threat landscape and lessons learned from past incidents.



Risk Management Communication

- ▶ Management of risk includes mitigating, transferring, avoiding, and accepting.
- ▶ Executives, managers and practitioners communicate with each other while to work on closing the gaps between the current and target profiles and address concerns that may arise.



Benefits of Adoption

- ▶ Enhance cybersecurity posture organization-wide
- ▶ Meet compliance requirements
- ▶ Safeguard organizational reputation
- ▶ Manage and minimize risk

Consequences of Non-Adoption

- ▶ Unable to keep up with evolving threats
- ▶ Inconsistencies with organization-wide risk management
- ▶ No verified way to measure effectiveness of cybersecurity controls
- ▶ Increased likelihood of incidents and longer recovery times

References

- ▶ The NIST Cybersecurity Framework (CSF) 2.0. (2024, February 26). *NIST*. <https://doi.org/10.6028/NIST.CSWP.29>
- ▶ Audit Peak. (2023, August 3). *Benefits and challenges in implementing NIST CSF*. AuditPeak. <https://www.auditpeak.com/challenges-in-implementing-nist-csf/>